

Question	Answer	Marks
1(a)	rate of change of velocity	B1
1(b)(i)	acceleration = $320 / 20$ $= 16 \text{ m s}^{-2}$	A1
1(b)(ii)	$s = \frac{1}{2} ((u) + v)t$ or distance = area under graph	C1
	(height) = $\frac{1}{2} \times 20 \times 320 = 3200 \text{ m}$ or 3.2 km	A1
1(c)(i)	$(\Delta E_P) = mg\Delta h$	C1
	$= 2.9 \times 10^6 \times 9.81 \times 3200$ $= 9.1 \times 10^{10} \text{ J}$	A1
1(c)(ii)	$(\Delta E_K) = \frac{1}{2} m\Delta(v^2)$	C1
	$= \frac{1}{2} \times 2.9 \times 10^6 \times 320^2$ $= 1.5 \times 10^{11} \text{ J}$	A1
1(c)(iii)	(power =) work done / time	C1
	power = $((1.5 + 0.91) \times 10^{11}) / 20$ $= 1.2 \times 10^{10} \text{ W}$	A1

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